4.1 Definition of the Laplace Transform

INTRODUCTION In elementary calculus you learned that differentiation and integration are *transforms*—this means, roughly speaking, that these operations transform a function into another function. For example, the function $f(x) = x^2$ is transformed, in turn, into a linear function, a family of cubic polynomial functions, and a constant by the operations of differentiation, indefinite integration, and definite integration:

$$\frac{d}{dx}x^2 = 2x, \quad \int x^2\,dx = \frac{1}{3}x^3 + c, \quad \int_0^3 x^2\,dx = 9.$$

Moreover, these two transforms possess the **linearity property**; this means the transform of a linear combination of functions is a linear combination of the transforms. For α and β constants,

$$\frac{d}{dx}[\alpha f(x) + \beta g(x)] = \alpha f'(x) + \beta g'(x)$$

$$\int [\alpha f(x) + \beta g(x)]\,dx = \alpha \int f(x)\,dx + \beta \int g(x)\,dx$$

and
$$\int_a^b [\alpha f(x) + \beta g(x)]\,dx = \alpha \int_a^b f(x)\,dx + \beta \int_a^b g(x)\,dx$$

provided each derivative and integral exists. In this section we will examine a special type of integral transform called the Laplace transform. In addition to possessing the linearity property, the Laplace transform has many other interesting properties that make it very useful in solving linear initial-value problems.

If $f(x, y)$ is a function of two variables, then a partial definite integral of f with respect to one of the variables leads to a function of the other variable. For example, by holding y constant we see that $\int_1^2 2xy^2\,dx = 3y^2$. Similarly, a definite integral such as $\int_a^b K(s, t)f(t)\,dt$ transforms a function $f(t)$ into a function of the variable s. We are particularly interested in **integral transforms** of this last kind, where the interval of integration is the unbounded interval $[0, \infty)$.

We will assume throughout that s is a real variable.

Basic Definition

If $f(t)$ is defined for $t \geq 0$, then the improper integral $\int_0^\infty K(s, t)f(t)\,dt$ is defined as a limit:

$$\int_0^\infty K(s, t)f(t)\,dt = \lim_{b\to\infty}\int_0^b K(s, t)f(t)\,dt. \tag{1}$$

If the limit exists, the integral is said to **exist** or to be **convergent**; if the limit does not exist, the integral does not exist and is said to be **divergent**. The foregoing limit will, in general, exist for only certain values of the variable s. The choice $K(s, t) = e^{-st}$ gives us an especially important integral transform.

Definition 4.1.1 Laplace Transform

Let f be a function defined for $t \geq 0$. Then the integral

$$\mathscr{L}\{f(t)\} = \int_0^\infty e^{-st}f(t)\,dt \tag{2}$$

is said to be the **Laplace transform** of f, provided the integral converges.

The Laplace transform most likely was invented by Leonhard Euler, but is named after the famous French astronomer and mathematician **Pierre-Simon Marquis de Laplace** (1749–1827) who used the transform in his investigations of probability theory.

When the defining integral (2) converges, the result is a function of s. In general discussion, if we use a lowercase (uppercase) letter to denote the function being transformed, then the corresponding uppercase (lowercase) letter will be used to denote its Laplace transform, for example,

$$\mathscr{L}\{f(t)\} = F(s), \quad \mathscr{L}\{g(t)\} = G(s), \quad \mathscr{L}\{y(t)\} = Y(s), \quad \text{and} \quad \mathscr{L}\{H(t)\} = h(s).$$

EXAMPLE 1 Using Definition 4.1.1

Evaluate $\mathscr{L}\{1\}$.

SOLUTION From (2),

$$\mathscr{L}\{1\} = \int_0^\infty e^{-st}(1)\,dt = \lim_{b\to\infty}\int_0^b e^{-st}\,dt$$

$$= \lim_{b\to\infty}\frac{-e^{-st}}{s}\bigg|_0^b = \lim_{b\to\infty}\frac{-e^{-sb}+1}{s} = \frac{1}{s}$$

provided $s > 0$. In other words, when $s > 0$, the exponent $-sb$ is negative and $e^{-sb} \to 0$ as $b \to \infty$. The integral diverges for $s < 0$. ≡

The use of the limit sign becomes somewhat tedious, so we shall adopt the notation $|_0^\infty$ as a shorthand to writing $\lim_{b\to\infty}(\ )\,|_0^b$. For example,

$$\mathscr{L}\{1\} = \int_0^\infty e^{-st}(1)\,dt = \frac{-e^{-st}}{s}\bigg|_0^\infty = \frac{1}{s}, \quad s > 0.$$

At the upper limit, it is understood we mean $e^{-st} \to 0$ as $t \to \infty$ for $s > 0$.

EXAMPLE 2 Using Definition 4.1.1

Evaluate $\mathscr{L}\{t\}$.

SOLUTION From Definition 4.1.1, we have $\mathscr{L}\{t\} = \int_0^\infty e^{-st}t\,dt$. Integrating by parts and using $\lim_{t\to\infty} te^{-st} = 0$, $s > 0$, along with the result from Example 1, we obtain

$$\mathscr{L}\{t\} = \frac{-te^{-st}}{s}\bigg|_0^\infty + \frac{1}{s}\int_0^\infty e^{-st}dt = \frac{1}{s}\mathscr{L}\{1\} = \frac{1}{s}\left(\frac{1}{s}\right) = \frac{1}{s^2}.$$ ≡

EXAMPLE 3 Using Definition 4.1.1

Evaluate **(a)** $\mathscr{L}\{e^{-3t}\}$ **(b)** $\mathscr{L}\{e^{6t}\}$.

SOLUTION In each case we use Definition 4.1.1.

(a) $$\mathscr{L}\{e^{-3t}\} = \int_0^\infty e^{-3t}e^{-st}dt = \int_0^\infty e^{-(s+3)t}dt$$

$$= \frac{-e^{-(s+3)t}}{s+3}\bigg|_0^\infty$$

$$= \frac{1}{s+3}.$$

The last result is valid for $s > -3$ because in order to have $\lim_{t\to\infty} e^{-(s+3)t} = 0$ we must require that $s + 3 > 0$ or $s > -3$.

(b) $\mathscr{L}\{e^{6t}\} = \int_0^\infty e^{6t}e^{-st}dt = \int_0^\infty e^{-(s-6)t}dt$

$$= \frac{-e^{-(s-6)t}}{s-6}\bigg|_0^\infty$$

$$= \frac{1}{s-6}.$$

In contrast to part (a), this result is valid for $s > 6$ because $\lim_{t\to\infty} e^{-(s-6)t} = 0$ demands $s - 6 > 0$ or $s > 6$. ≡

EXAMPLE 4 Using Definition 4.1.1

Evaluate $\mathscr{L}\{\sin 2t\}$.

SOLUTION From Definition 4.1.1 and integration by parts we have

$$\mathscr{L}\{\sin 2t\} = \int_0^\infty e^{-st}\sin 2t\,dt = \frac{-e^{-st}\sin 2t}{s}\bigg|_0^\infty + \frac{2}{s}\int_0^\infty e^{-st}\cos 2t\,dt$$

$$= \frac{2}{s}\int_0^\infty e^{-st}\cos 2t\,dt, \quad s > 0$$

$$\underset{\lim_{t\to\infty} e^{-st}\cos 2t \,=\, 0,\ s>0}{\downarrow} \qquad\qquad \underset{\text{Laplace transform of } \sin 2t}{\downarrow}$$

$$= \frac{2}{s}\left[\frac{-e^{-st}\cos 2t}{s}\bigg|_0^\infty - \frac{2}{s}\int_0^\infty e^{-st}\sin 2t\,dt\right]$$

$$= \frac{2}{s^2} - \frac{4}{s^2}\mathscr{L}\{\sin 2t\}.$$

At this point we have an equation with $\mathscr{L}\{\sin 2t\}$ on both sides of the equality. Solving for that quantity yields the result

$$\mathscr{L}\{\sin 2t\} = \frac{2}{s^2+4}, \qquad s > 0.$$ ≡

$\mathscr{L}$ Is a Linear Transform

For a sum of functions, we can write

$$\int_0^\infty e^{-st}[\alpha f(t) + \beta g(t)]\,dt = \alpha\int_0^\infty e^{-st}f(t)\,dt + \beta\int_0^\infty e^{-st}g(t)\,dt$$

whenever both integrals converge for $s > c$. Hence it follows that

$$\mathscr{L}\{\alpha f(t) + \beta g(t)\} = \alpha\mathscr{L}\{f(t)\} + \beta\mathscr{L}\{g(t)\} = \alpha F(s) + \beta G(s). \tag{3}$$

Because of the property given in (3), $\mathscr{L}$ is said to be a **linear transform**. Furthermore, by the properties of the definite integral, the transform of any finite linear combination of functions $f_1(t)$, $f_2(t), \ldots, f_n(t)$ is the sum of the transforms provided each transform exists on some common interval on the s-axis.

EXAMPLE 5 Linearity of the Laplace Transform

In this example we use the results of the preceding examples to illustrate the linearity of the Laplace transform.

(a) From Examples 1 and 2 we know that both $\mathscr{L}\{1\}$ and $\mathscr{L}\{t\}$ exist on the interval defined by $s > 0$. Hence, for $s > 0$ we can write

$$\mathscr{L}\{1 + 5t\} = \mathscr{L}\{1\} + 5\mathscr{L}\{t\} = \frac{1}{s} + \frac{5}{s^2}.$$

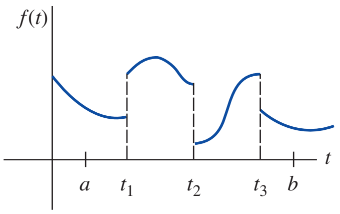

FIGURE 4.1.1 Piecewise-continuous function

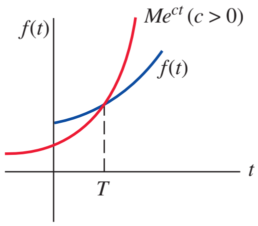

FIGURE 4.1.2 Function f is of exponential order

A more extensive list of transforms is given in Appendix III.

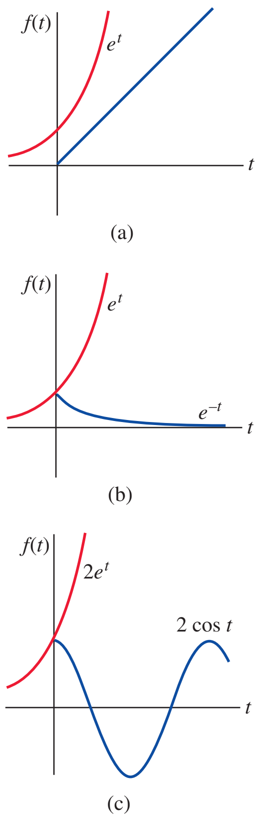

FIGURE 4.1.3 Functions with blue graphs are of exponential order

(b) From Example 3 we saw that $\mathscr{L}\{e^{6t}\}$ exists on the interval defined by $s > 6$, and in Example 4 we saw that $\mathscr{L}\{\sin 2t\}$ exists on the interval defined by $s > 0$. Thus both transforms exist for the common values of s defined by $s > 6$, and we can write

$$\mathscr{L}\{2e^{6t} - 15\sin 2t\} = 2\,\mathscr{L}\{e^{6t}\} - 15\,\mathscr{L}\{\sin 2t\} = \frac{2}{s-6} - \frac{15}{s^2+4}.$$

(c) From Examples 1, 2, and 3 we have for $s > 0$,

$$\mathscr{L}\{10e^{-3t} - 5t + 8\} = 10\,\mathscr{L}\{e^{-3t}\} - 5\mathscr{L}\{t\} + 8\,\mathscr{L}\{1\}$$
$$= \frac{10}{s+3} - \frac{5}{s^2} + \frac{8}{s}.$$

≡

We state the generalization of some of the preceding examples by means of the next theorem. From this point on we shall also refrain from stating any restrictions on s; it is understood that s is sufficiently restricted to guarantee the convergence of the appropriate Laplace transform.

Theorem 4.1.1 Transforms of Some Basic Functions

(a) $\mathscr{L}\{1\} = \dfrac{1}{s}$

(b) $\mathscr{L}\{t^n\} = \dfrac{n!}{s^{n+1}},\ n = 1, 2, 3, \ldots$ **(c)** $\mathscr{L}\{e^{at}\} = \dfrac{1}{s-a}$

(d) $\mathscr{L}\{\sin kt\} = \dfrac{k}{s^2+k^2}$ **(e)** $\mathscr{L}\{\cos kt\} = \dfrac{s}{s^2+k^2}$

(f) $\mathscr{L}\{\sinh kt\} = \dfrac{k}{s^2-k^2}$ **(g)** $\mathscr{L}\{\cosh kt\} = \dfrac{s}{s^2-k^2}$

Sufficient Conditions for Existence of $\mathscr{L}\{f(t)\}$

The integral that defines the Laplace transform does not have to converge. For example, neither $\mathscr{L}\{1/t\}$ nor $\mathscr{L}\{e^{t^2}\}$ exists. Sufficient conditions guaranteeing the existence of $\mathscr{L}\{f(t)\}$ are that f be piecewise continuous on $[0, \infty)$ and that f be of exponential order for $t > T$. Recall that a function f is piecewise continuous on $[0, \infty)$ if, in any interval defined by $0 \le a \le t \le b$, there are at most a finite number of points t_k, $k = 1, 2, \ldots, n$ $(t_{k-1} < t_k)$, at which f has finite discontinuities and is continuous on each open interval defined by $t_{k-1} < t < t_k$. See **FIGURE 4.1.1**. The concept of exponential order is defined in the following manner.

Definition 4.1.2 Exponential Order

A function f is said to be of **exponential order** if there exist constants c, $M > 0$, and $T > 0$ such that $|f(t)| \le Me^{ct}$ for all $t > T$.

If f is an *increasing* function, then the condition $|f(t)| \le Me^{ct}$, $t > T$, simply states that the graph of f on the interval (T, ∞) does not grow faster than the graph of the exponential function Me^{ct}, where c is a positive constant. See **FIGURE 4.1.2**. The functions $f(t) = t$, $f(t) = e^{-t}$, and $f(t) = 2\cos t$ are all of exponential order $c = 1$ for $t > 0$ since we have, respectively,

$$|t| \le e^t, \quad |e^{-t}| \le e^t, \quad |2\cos t| \le 2e^t.$$

A comparison of the graphs on the interval $[0, \infty)$ is given in **FIGURE 4.1.3**.

A positive integral power of t is always of exponential order since, for $c > 0$,

$$|t^n| \le Me^{ct} \quad \text{or} \quad \left|\frac{t^n}{e^{ct}}\right| \le M \quad \text{for } t > T$$

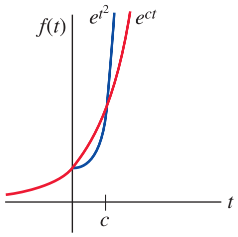

FIGURE 4.1.4 $f(t) = e^{t^2}$ is not of exponential order

is equivalent to showing that $\lim_{t\to\infty} t^n/e^{ct}$ is finite for $n = 1, 2, 3, \ldots$. The result follows by n applications of L'Hôpital's rule. A function such as $f(t) = e^{t^2}$ is not of exponential order since, as shown in **FIGURE 4.1.4**, e^{t^2} grows faster than any positive linear power of e for $t > c > 0$. This can also be seen from

$$\left|\frac{e^{t^2}}{e^{ct}}\right| = e^{t^2 - ct} = e^{t(t-c)} \to \infty \text{ as } t \to \infty$$

for *any* value of c. By the same reasoning, $e^{-st}e^{t^2} \to \infty$ as $t \to \infty$ for any s and so the improper integral $\int_0^\infty e^{-st}e^{t^2}\,dt$ diverges. In other words, $\mathscr{L}\{e^{t^2}\}$ does not exist.

Theorem 4.1.2 Sufficient Conditions for Existence

If $f(t)$ is piecewise continuous on the interval $[0, \infty)$ and of exponential order, then $\mathscr{L}\{f(t)\}$ exists for $s > c$.

PROOF: By the additive interval property of definite integrals,

$$\mathscr{L}\{f(t)\} = \int_0^T e^{-st}f(t)\,dt + \int_T^\infty e^{-st}f(t)\,dt = I_1 + I_2.$$

The integral I_1 exists because it can be written as a sum of integrals over intervals on which $e^{-st}f(t)$ is continuous. Now f is of exponential order, so there exists constants c, $M > 0$, $T > 0$ so that $|f(t)| \le Me^{ct}$ for $t > T$. We can then write

$$|I_2| \le \int_T^\infty |e^{-st}f(t)|\,dt \le M\int_T^\infty e^{-st}e^{ct}\,dt = M\int_T^\infty e^{-(s-c)t}\,dt = M\frac{e^{-(s-c)T}}{s-c}$$

for $s > c$. Since $\int_T^\infty Me^{-(s-c)t}\,dt$ converges, the integral $\int_T^\infty |e^{-st}f(t)|\,dt$ converges by the comparison test for improper integrals. This, in turn, implies that I_2 exists for $s > c$. The existence of I_1 and I_2 implies that $\mathscr{L}\{f(t)\} = \int_0^\infty e^{-st}f(t)\,dt$ exists for $s > c$. ≡

EXAMPLE 6 Transform of a Piecewise-Continuous Function

Evaluate $\mathscr{L}\{f(t)\}$ for $f(t) = \begin{cases} 0, & 0 \le t < 3 \\ 2, & t \ge 3. \end{cases}$

SOLUTION This piecewise-continuous function appears in **FIGURE 4.1.5**. Since f is defined in two pieces, $\mathscr{L}\{f(t)\}$ is expressed as the sum of two integrals:

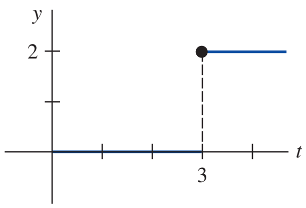

FIGURE 4.1.5 Piecewise-continuous function in Example 6

$$\begin{aligned}\mathscr{L}\{f(t)\} &= \int_0^\infty e^{-st}f(t)\,dt = \int_0^3 e^{-st}(0)\,dt + \int_3^\infty e^{-st}(2)\,dt \\ &= -\frac{2e^{-st}}{s}\Bigg|_3^\infty \\ &= \frac{2e^{-3s}}{s}, \quad s > 0.\end{aligned}$$

≡

REMARKS

Throughout this chapter we shall be concerned primarily with functions that are both piecewise continuous and of exponential order. We note, however, that these two conditions are sufficient but not necessary for the existence of a Laplace transform. The function $f(t) = t^{-1/2}$ is not piecewise continuous on the interval $[0, \infty)$; nevertheless its Laplace transform exists. See Problem 43 in Exercises 4.1.

4.1 Exercises Answers to selected odd-numbered problems begin on page ANS-8.

In Problems 1–18, use Definition 4.1.1 to find $\mathscr{L}\{f(t)\}$.

1. $f(t) = \begin{cases} -1, & 0 \le t < 1 \\ 1, & t \ge 1 \end{cases}$

2. $f(t) = \begin{cases} 4, & 0 \le t < 2 \\ 0, & t \ge 2 \end{cases}$

3. $f(t) = \begin{cases} t, & 0 \le t < 1 \\ 1, & t \ge 1 \end{cases}$

4. $f(t) = \begin{cases} 2t + 1, & 0 \le t < 1 \\ 0, & t \ge 1 \end{cases}$

5. $f(t) = \begin{cases} \sin t, & 0 \le t < \pi \\ 0, & t \ge \pi \end{cases}$

6. $f(t) = \begin{cases} \sin t, & 0 \le t < \pi/2 \\ 0, & t \ge \pi/2 \end{cases}$

7.

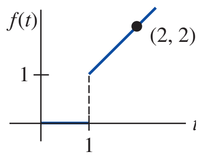

FIGURE 4.1.6 Graph for Problem 7

8.

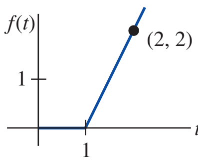

FIGURE 4.1.7 Graph for Problem 8

9.

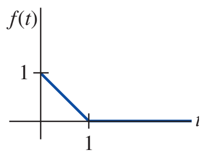

FIGURE 4.1.8 Graph for Problem 9

10.

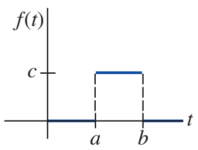

FIGURE 4.1.9 Graph for Problem 10

11. $f(t) = e^{t+7}$
12. $f(t) = e^{-2t-5}$
13. $f(t) = te^{4t}$
14. $f(t) = t^2e^{-2t}$
15. $f(t) = e^{-t}\sin t$
16. $f(t) = e^t \cos t$
17. $f(t) = t\cos t$
18. $f(t) = t\sin t$

In Problems 19–36, use Theorem 4.1.1 to find $\mathscr{L}\{f(t)\}$.

19. $f(t) = 2t^4$
20. $f(t) = t^5$
21. $f(t) = 4t - 10$
22. $f(t) = 7t + 3$
23. $f(t) = t^2 + 6t - 3$
24. $f(t) = -4t^2 + 16t + 9$
25. $f(t) = (t + 1)^3$
26. $f(t) = (2t - 1)^3$
27. $f(t) = 1 + e^{4t}$
28. $f(t) = t^2 - e^{-9t} + 5$
29. $f(t) = (1 + e^{2t})^2$
30. $f(t) = (e^t - e^{-t})^2$
31. $f(t) = 4t^2 - 5\sin 3t$
32. $f(t) = \cos 5t + \sin 2t$
33. $f(t) = \sinh kt$
34. $f(t) = \cosh kt$
35. $f(t) = e^t \sinh t$
36. $f(t) = e^{-t}\cosh t$

In Problems 37–40, find $\mathscr{L}\{f(t)\}$ by first using an appropriate trigonometric identity.

37. $f(t) = \sin 2t \cos 2t$
38. $f(t) = \cos^2 t$
39. $f(t) = \sin(4t + 5)$
40. $f(t) = 10\cos(t - \pi/6)$

41. One definition of the **gamma function** $\Gamma(\alpha)$ is given by the improper integral

$$\Gamma(\alpha) = \int_0^\infty t^{\alpha-1}e^{-t}dt, \quad \alpha > 0.$$

Use this definition to show that $\Gamma(\alpha + 1) = \alpha\Gamma(\alpha)$.

42. Use Problem 41 to show that

$$\mathscr{L}\{t^\alpha\} = \frac{\Gamma(\alpha + 1)}{s^{\alpha+1}}, \quad \alpha > -1.$$

This result is a generalization of Theorem 4.1.1(b).

In Problems 43–46, use the results in Problems 41 and 42 and the fact that $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ to find the Laplace transform of the given function.

43. $f(t) = t^{-1/2}$
44. $f(t) = t^{1/2}$
45. $f(t) = t^{3/2}$
46. $f(t) = 6t^{1/2} - 24t^{5/2}$

Discussion Problems

47. Suppose that $\mathscr{L}\{f_1(t)\} = F_1(s)$ for $s > c_1$ and that $\mathscr{L}\{f_2(t)\} = F_2(s)$ for $s > c_2$. When does

$$\mathscr{L}\{f_1(t) + f_2(t)\} = F_1(s) + F_2(s)?$$

48. Figure 4.1.4 suggests, but does not prove, that the function $f(t) = e^{t^2}$ is not of exponential order. How does the observation that $t^2 > \ln M + ct$, for $M > 0$ and t sufficiently large, show that $e^{t^2} > Me^{ct}$ for any c?

49. Use part (c) of Theorem 4.1.1 to show that

$$\mathscr{L}\{e^{(a+ib)t}\} = \frac{s - a + ib}{(s - a)^2 + b^2},$$

where a and b are real and $i^2 = -1$. Show how Euler's formula (page 121) can then be used to deduce the results

$$\mathscr{L}\{e^{at}\cos bt\} = \frac{s - a}{(s - a)^2 + b^2}$$

and
$$\mathscr{L}\{e^{at}\sin bt\} = \frac{b}{(s - a)^2 + b^2}.$$

50. Under what conditions is a linear function $f(x) = mx + b$, $m \ne 0$, a linear transform?

51. The function $f(t) = 2te^{t^2}\cos e^{t^2}$ is not of exponential order. Nevertheless, show that the Laplace transform $\mathscr{L}\{2te^{t^2}\cos e^{t^2}\}$ exists. [*Hint*: Use integration by parts.]

52. Explain why the function

$$f(t) = \begin{cases} t, & 0 \le t < 2 \\ 4, & 2 < t < 4 \\ 1/(t - 4), & t > 4 \end{cases}$$

is not piecewise continuous on $[0, \infty)$.